FINAL

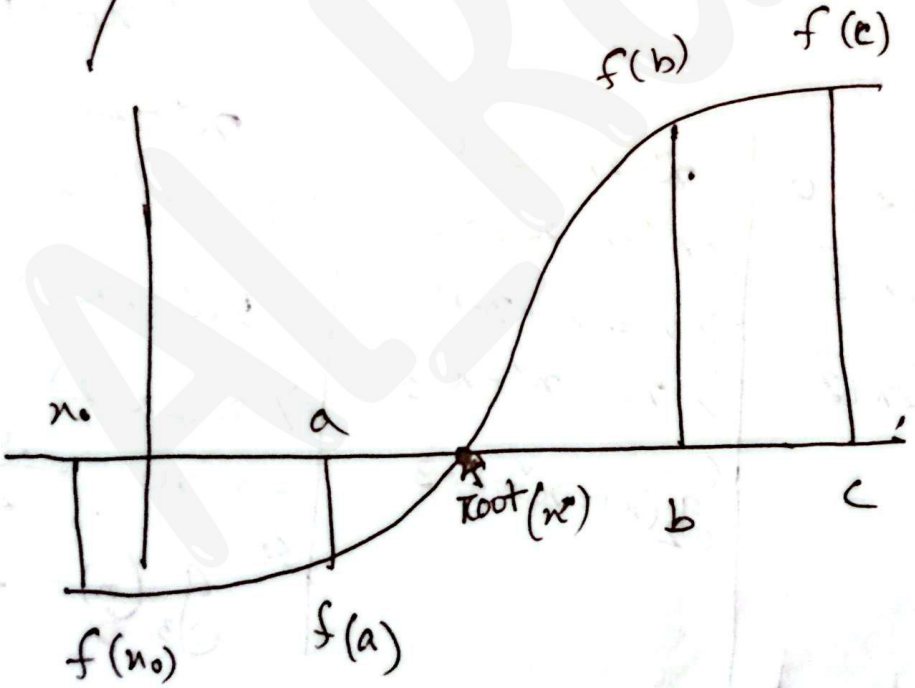
CHAPTER-4

Root Finding Algorithms

There are 2 methods.

① ~~Bisectio~~ ① Bracketed Method ✗

② Open / Non-Bracketed Method ✓



$f(x_0) \times f(a) > 0$

$f(b) \times f(c) > 0$

but $f(a) \times f(b) < 0$

Open Method

Root Finding Methods →

যেখানে curve যে point-এ x axis কে ছেদ করে.

① Fixed Point Iteration

① Fast

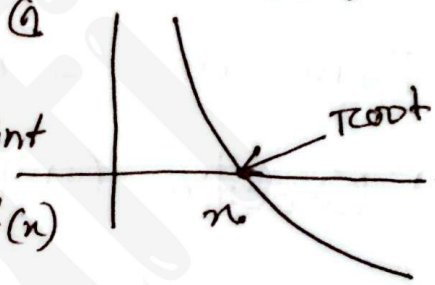
② Doesn't guarantee convergence.

$f(x) = 0$; এখানে $f(x)$ কে $g(x)$ এ সমানুপাত করব। then,

if $g(x) = x$; then, → fixed point

↳ root of the actual $f(x)$

root এ, $f(x) = 0$



$f(x) = x^2 - 2x - 3$; $f(x) = 0$; $x_0 = 0$

$x^2 - 2x + 3 = 0$

$\Rightarrow x = \sqrt{2x+3}$

So, $g_1(x) = \sqrt{2x+3}$

$x^2 - 2x - 3 = 0$

$\Rightarrow x^2 - x - x - 3 = 0$

$\Rightarrow -x = x + 3 - x^2$

$\Rightarrow x = x^2 - x - 3$

So, $g_2(x) = x^2 - x - 3$

$x^2 - 2x - 3 = 0$

$\Rightarrow 2x^2 - x^2 - 2x - 3 = 0$

$\Rightarrow 2x^2 - 2x = x^2 + 3$

$\Rightarrow x(2x-2) = x^2 + 3$

$\Rightarrow x = \frac{x^2+3}{2x-2}$

So, $g_3(x) = \frac{x^2+3}{2x-2}$

Here, Actual root,

$f(x) = 0$

$\Rightarrow x^2 - 2x + 3 = 0$

So, $x = -1, x = 3$

So, we get the actual roots, $x = 3$ & $x = -1$

Now, $\boxed{x_{n+1} = g(x_n)}$

$g_1(x) = \sqrt{2x+3}$; $x_0 = 0$

$x_0 = 0$; $g_1(x) = \sqrt{2x+3}$

So,

$x_0 = 0$; $g_1(0) =$

$x_1 = 1.73$; $g_1(1.73) = 2.54$

$x_2 = 2.54$; $g_1(2.54) = 2.84$

$x_3 = 2.84$; $g_1(2.84) = 2.95$

$x_4 = 2.95$; $g_1(2.95) = 2.98$

$x_5 = 2.98$; $g_1(2.98) = 3.0$

$x_6 = 3.0$; $g_1(3.0) = 3.0$

$x_7 = 3.0$;

Here, Root of $g_1(x)$; Root value ~~ই~~ সাত্ত্ব ~~ফিরে~~ পাচ্ছি ; value same.

So, It's convergent.

Now, for g_2,

$g_2(x) = x^2 - x - 3$

$g_2(0) = -3$

$g_2(-3) = 9.00$

$g_2(9.00) = 69.0$

No matching value ; So,

Divergent

Now for $g_3(x) = \frac{x^2+3}{2x-2}$; $x_0 = 0$

So,

$g_3(0) = -1.5$

$g_3(-1.5) = -1.05$

$g_3(-1.05) = -1.00$

$g_3(-1.00) = -1.0$

↳ Root of the $f(x)$

যেই x_0 এর মান বসে root এর value এর কাছে output এ সেই value পাব.

- কখন কোন Root পাব output টা initial point এর উপর depend করে.

Contraction Mapping Theorem

To find out if it converges or not.

$$\lambda = |g'(x)| = \begin{cases} \lambda = 0 \text{ ; Super Linear} \\ 0 < \lambda < 1 \text{ ; Linear} \end{cases} \Big\} \text{ Convergent}$$

$$\lambda \geq 1 \text{ ; Divergent}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

$f(x) = x^3 - 2x^2 - x + 2$

(a) Find the roots of $f(x)$

Let, $x^3 - 2x^2 - x + 2 = 0$

$\Rightarrow x^2(x-2) - 1(x-2) = 0$

$\Rightarrow (x^2-1)(x-2) = 0$

So, $x = 1$, $x = -1$, $x = 2$

(b) Construct 3 $g(x)$ for $f(x)$

$x^3 - 2x^2 - x + 2$

$x^3 - 2x^2 = x - 2$

$x = x^3 - 2x^2 + 2$

$x^3 - 2x^2 - x + 2$

$x^3 - x + 2 = 2x^2$

$\Rightarrow x^2 = \frac{1}{2}(x^3 - x + 2)$

$\therefore x = \sqrt{\frac{1}{2}(x^3 - x + 2)}$

$x^3 - 2x^2 - x + 2$

$x = 3\sqrt{2x^3 + x - 2}$

$\Rightarrow x^3 = 2x^2 + x - 2$

$x = \frac{-2}{x^2 - 2x - 1}$

So, we get,

$$g_1(x) = \sqrt{\frac{1}{2}(x^3 - x + 2)} \qquad g_3(x) = \frac{-2}{x^2 - 2x - 1}$$

$$g_2(x) = x^3 - 2x^2 + 2$$

(c) Determine which $g(x)$ are convergent or divergent.

We know,

$$\lambda = |g_i'(x)|$$

for, $g_1(x) = \sqrt{\frac{1}{2}(x^3 - x + 2)}$

So, $g_1'(x) = \frac{1}{\sqrt{2}} \cdot (x^3 - x + 2)^{\frac{1}{2}}$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{2}(x^3 - x + 2)^{-\frac{1}{2}} \cdot (3x^2 - 1)$$

$$= \frac{3x^2 - 1}{2\sqrt{2}(x^3 - x + 2)^{\frac{1}{2}}}$$

So, $\lambda = |g_1'(x)| = \left| \frac{3x^2 - 1}{2\sqrt{2}(x^3 - x + 2)^{1/2}} \right|$

Roots are , 1, -1, 2

So,

$|g_1'(1)| = 0.5$ Linear | $g_1'(-1) = 0.5$ Linear | $g_1'(2) = 1.375$ Divergent

Here, $g_2(x) = x^3 - 2x^2 + 2$

So, $g_2'(x) = 3x^2 - 4x$

So,

$\lambda = |g_2'(x)| = |3x^2 - 2x|$

- $x=1$; 1 → Div
- $x=-1$; 5 → Div
- $x=2$; 4 → Div

Again $g_3(x) = \frac{-2}{x^2-2x-1} = -2(x^2-2x-1)^{-1}$

So, $g_3'(x) = 2(x^2-2x-1)^{-2}(2x-2) = \frac{2(2x-2)}{(x^2-2x-1)^2}$

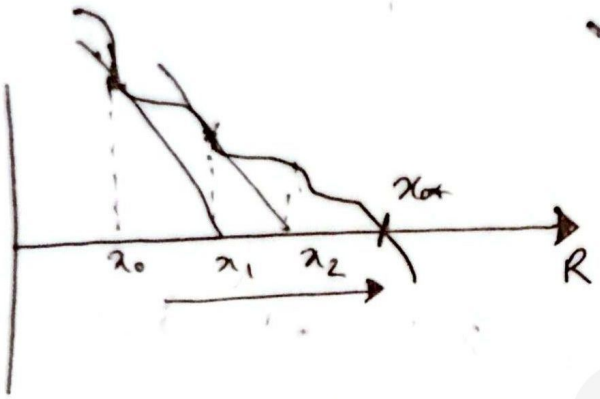
So,

$\lambda = \left|g_3'(x)\right| = \left|\frac{2(2x-2)}{(x^2-2x-1)^2}\right|$

- $x=1$; 0 → Converges Superlinear
- $x=-1$; 2 → Div
- $x=2$; 4 → Div

Newton's Rhapson Method

↳ Super linear convergence



Formula

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Ex → $f(x) = x^2 - 2xe^{-x} + e^{-2x}$; $x_0 = 1$; Error Bound $= 1 \times 10^{-5}$

k	x_k	$f(x_k)$	$f(x_k) <$ Error Bound
0	1	0.3996	No
1	0.7683	0.093	No
2	0.6648	0.0226	No
⋮	⋮	⋮	⋮
8	0.56	0.49×10^{-5}	Yes

Here,

$f(x) = x^2 - 2xe^{-x} + e^{-2x}$

So, $f'(x) = 2x - [2e^{-x} + 2xe^{-x}(-1) + -2e^{-2x}]$

$= 2x - 2e^{-x} + 2xe^{-x} - 2e^{-2x}$

So,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1 - \frac{0.3996}{1.7293}$$

$$= 0.7683$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.7689 - \frac{f(0.7689)}{f'(0.7689)}$$

$$= 0.6648$$

So, 0.56 is the root of a function.

(Ans.)

Ex → $f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$; $x_0 = 0.05$

Show first 3 iteration with relative errorr at each iteration.

Here,

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

So, $f'(x) = 3x^2 - 0.33x$

Iteration-1

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.05 - \frac{f(0.05)}{f'(0.05)}$$

$$= 0.05 - \frac{1.118 \times 10^{-4}}{-9 \times 10^{-3}}$$

$$= 0.0624$$

R.E. $= \left| x_{New} - x_{old} \right| = |0.0624 - 0.05| = 0.0124$

Iteration - 2

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.0624 - \frac{f(0.0624)}{f'(0.0624)}$$

$$= 0.0623$$

$$Rf = |x_2 - x_1| = |0.0623 - 0.0624|$$

$$= 0.0001$$

Iteration - 3

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.0001 - \frac{f(0.0001)}{f'(0.0001)}$$

$$= 0.0623$$

So, $Rf = |x_3 - x_2| = |0.0623 - 0.0001| = 0.0622$

Proving that it's superlinear Convergent

we know, $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$

So, $g(x) = x - \frac{f(x)}{f'(x)}$

$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$

$= \frac{u'v - uv'}{v^2}$

$\lambda = |g'(x)| \rightarrow \lambda, \lambda, \lambda$ ... x_{α}, x_{α}, x_{α}

$g(x) = x - f(x)/f'(x)$

$$\lambda = |g'(x)| = \left[1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2}\right]$$

$$= \frac{f(x)f''(x)}{[f'(x)]^2}$$

Let, x_α is my root.

So, $$\lambda = |g'(x_\alpha)| = \left| \frac{\cancel{f(x_\alpha)}^{0} f''(x_\alpha)}{[f'(x_\alpha)]^2} \right|$$

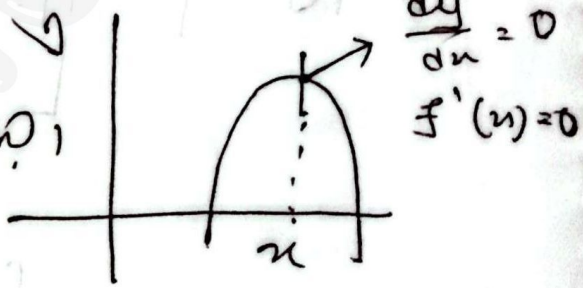
Super linear Method ও $\lambda = 0$ হবে।

$f(x_*) = 0$ হলে super linear হয় যদি x_* যে f(x) এর জন্য যদি value 0 হয়

Disadvantage

① $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$; $f'(x_k) = 0$ হলে undifined হয়।

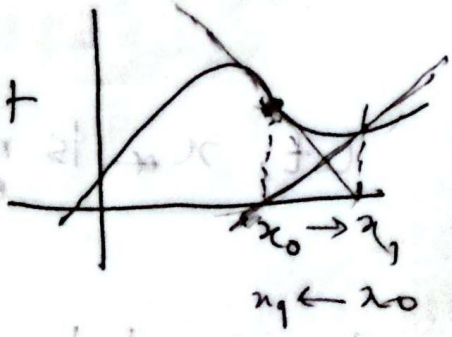
iterate করতে করতে x এর value tunning point এ reach করলে $f'(x) = 0$ হয়।



②

Infinite loop এ পড়ে যায়।

initial point যদি tunning point এর আশে পাশে থাকে।



③ $f'(x)$ is difficult to calculate.

$x=0$

Ex → $f(x) = 2x^2 - x - 2$; $x_0 = 4$;

Find a root accurate upto 3 decimal places.

Here, $f(x) = 2x^2 - x - 2$; $f'(x) = 4x - 1$

	k	x_k	$f(x_k)$
x_0	0	4	26
x_1	1	2.2667	6.0094
x_2	2	1.5218	1.1110
x_3	3	1.3036	0.0352
x_4	4	1.2810	0.0009

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 4 - \frac{f(4)}{f'(4)}$$

$$= 4 - \frac{26}{15} = 2.2667$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2.2667 - \frac{6.0094}{8.0668}$$

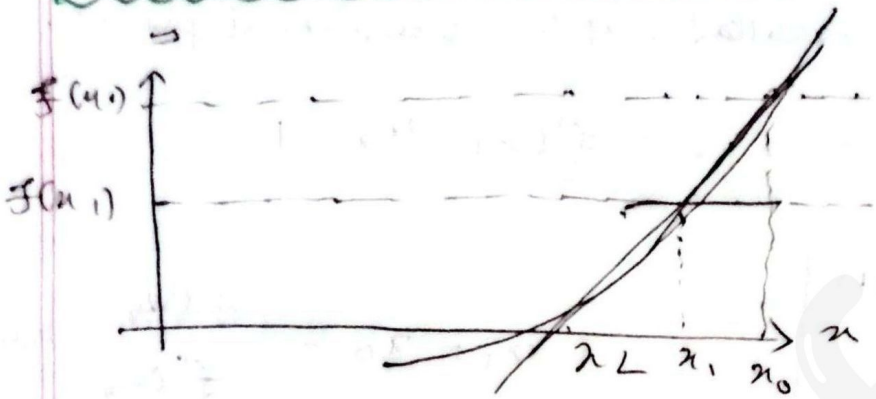
$$= 2.2667 - \frac{f(2.2667)}{f'(2.2667)}$$

$$= 1.5218$$

So, Approx Root, $x_\alpha = 1.2810$

(Ans)

Secant / Quasi Newton Method



$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{f(x_n)}{\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}}$$

$$= \frac{x_n - f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$f'(x_1) \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$f'(x_n) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

Formula

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Benefits

① It is always superlinear & better than NR Method as no need to find derivative.

② Here, divided by 0 issue নাই।

Ex → $f(x) = 2x^3 - x - 2$; $x_0 = 4$; $x_1 = 2$

Here, $f(x) = 2x^3 - x - 2$

x_{n-1}	x_n	x_{n+1}	$f(x_{n-1})$	$f(x_n)$	$f(x_{n+1})$
4	2	1.6364	26	4	1.7192
2	1.6364	1.3623	4	1.7192	0.3494
1.6364	1.3623		1.7192	0.3494	

Easier Alternative to Prev. method

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = x_2 - \frac{f(x_1)(x_2 - x_0)}{f(x_1) - f(x_0)} = 2 - \frac{4(2)}{4 - 26}$$